

Mainstream aberration / Airy derivations — project_sources index

PDFs live here in project_sources/. Full parsed source notes (screenshots, page-cited quotes, math, contextualized cards) live in the Obsidian vault /Sources/ and are backlinked from the Airy master index Airy/00_index.md (amendment 2026-07-10).

The question these sources answer: **does any mainstream derivation produce the aberration reciprocally, 20" in the Sun frame and 20" in the Earth frame?** None do. Each obtains 20.5" only in the observer's rest frame; the moving frame requires partial drag and yields the smaller drag family ($\beta/n \approx 15''$, $\beta(n-1/n) \approx 12''$), and each source states the angle is different in the two frames.

Parsed (full vault notes done)

PDF (project_sources)	Vault note	Key pages	What it shows / where it breaks
1921_Pauli_Theory_of_Relativity.pdf	Sources/1921_Pauli_Theory_of_Relativity.pdf	§36 (v/c) p. 14, §36 pp. 15-20	And "self-evident" only "from the rest system"; moving frame needs "rather involved arguments"; relations (311b,c) "do not correspond to the addition theorem... only for $\alpha=0$ " (plane-wave vs ray-velocity split)
1964_Rosser_Introduction_to_Science_The_1964_Relativistic_Hypothesis.pdf	Sources/1964_Rosser_Introduction_to_Science_The_1964_Relativistic_Hypothesis.pdf	§4.4.5 p. 154; §4.4.6 p. 157	SR aberration md $\Delta\alpha=(v/c)=20.5''$; Airy resolved entirely in the geocentric frame (water at rest, normal incidence, no refraction); $\theta' \neq \theta$ between frames; ray $\neq$ wave in a medium except parallel to motion (Møller)
1911_Watson_Textbook_of_Physics_1911_Light_and_Color.pdf	Sources/1911_Watson_Textbook_of_Physics_1911_Light_and_Color.pdf	§36 p. 507, §36 pp. 507-508	Standard textbook: water "should" increase the constant; "no such effect is, however, observed"; calls it "the difficulty," declines to resolve
1871_Airy_Supposed_Aberration_1871_Aberration_Supposed.pdf	Sources/1871_Aberration_Supposed.pdf	§20, §21	Aberration result being explained
1851_Fizeau_Hypothesis_of_Luminiferous_Ether_1851_Fizeau_Hypothesis.pdf	Sources/1851_Fizeau_Hypothesis_of_Luminiferous_Ether_1851_Fizeau_Hypothesis.pdf	§34	Fresnel drag coefficient $f = 1-1/n^2$

PDF (project_sources)	Vault note	Key pages	What it shows / where it breaks
2021_Gift_Simple_Test_Sources	2021_Gift_Simple_Test_Sources	JAMP 9:260-268, Eqs. 14-16	Let Drag drag gives $\Delta t = 2Lv/c^2$ (independent of n) for counter-propagating beams; SR predicts null; Wang et al. measured it → SR “cannot be a correct physical theory,” preferred frame = ECI/GPS
2510.20695_FromDragToInvariance.pdf	2025_Einstein's Invariance	arXiv:2510.20695, pp. 10-11	mainstream premise stated as principle: “aberration is not an internal flight-time effect... fixed at the moment of entry”; sources Boscovich 1766 + $\tan \alpha = v\phi/v_{\text{light}}$ (Eq. 37)
2014_WaterFilledTelescope	2014_Antonello's Water-Filled Telescope	arXiv:1401.5585, pp. 1-3	Wilspe's correction: with refraction included, corpuscular theory ALSO gives no change; the null is over-determined (geometric before relativistic) → zero discriminating power

Notes

- **DOI (Gift 2021):** 10.4236/jamp.2021.911171 · **ISSN** 2327-4379. Cites Wang, R. et al. (2003), *Modified Sagnac Experiment for Measuring Travel-Time Difference*.
- Weinstein and Antonello are open-access arXiv (physics.hist-ph). Gift is open-access (SCIRP / JAMP).

Verification companions (lighter, in this folder)

- 2025_Ligabue_Airy_Relativity.md — Franco's slides pinned to eq/slide numbers (his 15" Sun rest, 12" Sun moving)
- 1905_Einstein_Elektrodynamik_bewegter_Koerper.md — §7 aberration formula (0 in the shared rest frame)